

Practice Question Paper

SE303T – Software Construction & Development

Time Allowed: 2 Hours

Total Marks: 50

(For Practice & Preparation Only)

Instructions

- This is a practice paper to help you prepare.
- Focus on conceptual understanding and reasoning.
- Answers should explain why, not just what.

PART A - Theoretical (25 Marks)

Attempt ALL questions.

(Each question carries 5 marks)

Q1. A software team documents requirements using user stories and skips UML or BPMN models.

- Explain two risks of this approach
- Describe one situation where UML or BPMN would significantly improve understanding

Q2. A system is designed as a single large service because “it is easier to deploy.”

- Explain why this decision may cause scalability and maintenance issues
- How would modularity or architectural patterns help?

Q3. An ETL pipeline works correctly but becomes slow as data volume increases.

- Give two architectural reasons (not code-level bugs) for this issue
- Explain how architectural thinking helps identify the root cause

Q4. A team relies only on dashboards to monitor their system.

- Explain why dashboards alone are insufficient
- What additional observability practices should be included?

Q5. An engineering team frequently changes tools instead of improving system design.

- Explain how this behavior affects long-term system quality
- How does architectural evaluation help avoid this trap?

PART B - Scenario-Based Design (25 Marks)

Attempt ALL questions.

(Each question carries 5 marks)

Scenario:

You are designing a data processing pipeline for an organization that collects data from multiple departments. Some data arrives late, some is incomplete, and formats vary.

Q1. How would you design the pipeline to handle incomplete or delayed data without losing important information? Explain your reasoning.

Q2. Explain how you would ensure that each transformation step is traceable so that future audits can understand how data changed.

Q3. The organization plans to add real-time data processing in the future. What design constraints would you enforce now to support this future change?

Q4. Duplicate data starts appearing due to system integration issues. Explain how your pipeline would detect logical duplication, not just identical rows.

Q5. Your processed data is used to train an ML model. Explain two design mistakes that could make the model appear accurate but fail in real-world usage.